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Inhomogeneous shearing deformation of a rubber-like slab within the context of finite thermoelasticity with entropic origin for the stress

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Abstract

This paper deals with inhomogeneous shearing deformation of an infinite rubber-like slab under a temperature difference across its thickness. The deformation is analyzed within the context of finite thermoelasticity with entropic origin for the stress. The isothermal strain energy functions of Mooney and of Yeoh are generalized by incorporating them into the thermoelastic model. The energy balance equation, which is decoupled from the linear momentum balance equations, is solved for the temperature field using the Fourier's law of heat conduction. The derived linear temperature field and the equations of the thermoelastic model are inserted into the linear momentum balance equations, thus yielding an ordinary differential equation for the shear strain. The shear strain, the degree of inhomogeneity in the shear strain, the stress field, and the principal stresses are then determined using a finite difference method. The Mooney slab undergoes markedly greater shear strain near the colder boundary, thus exhibiting a boundary layer-like structure at higher temperature differences across the thickness. However, the Yeoh slab exhibits a relatively smaller variation of the shear strain even at the highest temperature difference. This difference in the behavior of the two slabs is attributed to the shear stiffening of the Yeoh slab, which counteracts the formation of a boundary layer-like structure. Both slabs experience constant shear stresses across the thickness, whereas they experience much greater normal stress differences varying across the thickness. The values of the major principal stress turn out to be close to those of the first normal stress difference. The presence of a large normal stress difference compared with a relatively smaller shear stress in the slabs creates the possibility for Mode I type fracture propagation.

Keywords: Inhomogeneous shearing; Finite thermoelasticity; Finite difference technique

1. Introduction

The success or failure of rubber-like materials in many practical applications depends heavily on the understanding of the thermoelastic response of these materials when subjected to temperature gradients and finite deformations. Rubber-like materials show non-linear stress–strain behavior, exhibit large deformation even under small loads, and have material parameters dependent on temperature [1]. The coupling between the deformation and temperature increases the complexity of the non-linear equations that arise in finite deformation of rubber-like materials.

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The earliest thermoelastic model in the literature is called the Gaussian model, which generalizes the isothermal neo-Hookean model by taking the modulus to be a linear function of absolute temperature. However, it is well known that the neo-Hookean model does not successfully predict even isothermal rubber-like behavior. One of the thermoelastic models intended to improve on the Gaussian model was developed by Chadwick [2]. In fact, what Chadwick developed is a phenomenological framework to determine a class of Helmholtz free energy functions that describe the behavior of isotropic thermoelastic materials. Few boundary value problems, however, have been solved within the context of Chadwick's model. Although many universal controllable deformations have been investigated by researchers, e.g. [3,4], few truly inhomogeneous deformations have been studied even within the context of isothermal conditions. It is well known that only homogeneous deformations are possible in every isotropic compressible elastic material [5]. In addition to these, universal solutions for isotropic incompressible elastic materials include five families of special inhomogeneous deformations [6]. Hence, in general, an inhomogeneous deformation should be studied within the context of some subclass of isotropic elastic materials.

Recently, many researchers have been interested in inhomogeneous deformations of some subclass of isotropic materials and have shown the existence of boundary layer-like structures (BLSs) in the deformation field [7–13]. A BLS occurs adjacent to a boundary where the strains are essentially inhomogeneous and much higher than the strains elsewhere in the deformed body. Such inhomogeneous solutions can arise from material non-linearities including temperature dependency of the material parameters in the model. Inhomogeneous shear deformation of a generalized neo-Hookean slab under a temperature gradient and several pressure gradients along the shearing direction has been investigated by Rajagopal [8]. It has been shown that the temperature and the stretch dependence of the material moduli can result in either accentuation or annihilation of the BLS. In the present work, we investigate the shearing deformation of a rubber-like slab subjected to a temperature gradient and prescribed displacement boundary conditions within the context of finite thermoelasticity with entropic origin for the stress, which is a subset of the thermoelastic model developed by Chadwick [2]. A major goal of this study is to determine the conditions under which a BLS can form in the sheared slab. The isothermal strain energy functions of Mooney [14] and of Yeoh [15] are generalized by incorporating them into the thermoelastic model. After solving for the decoupled temperature field analytically using the energy balance equation, linear momentum balance equations including temperature as a known linear function are solved for the shear strain. Our interest also lies in the determination of the stress field, the principal stresses, and the principal directions that seem to be instrumental in possible Mode I type fracture propagation in the slab. We observe that the generalized Mooney and the Yeoh slabs have substantially different stress–strain behavior due to the shear stiffening of the Yeoh slab.

2. Kinematical description of inhomogeneous shear

In this section, we postulate an inhomogeneous shear deformation and a temperature field, and give the measures of finite deformation. Let us consider an infinite slab bounded by the planes $Y = 0$ and H (where H is the thickness of the slab) in the undeformed state referred to the Cartesian coordinate system (X, Y, Z). The slab is assumed to be an isotropic incompressible elastic material whose stress–strain behavior is described by a finite thermoelastic model with entropic origin for the stress, as will be explained in the next section of this paper. The slab is subjected to a temperature difference across its thickness and sheared in the X direction by prescribed boundary displacements at $Y = 0$ and H. We restrict ourselves to the steady-state shearing of the slab.

Let $\mathbf{X} = (X, Y, Z)$ denote a material point of the reference configuration of the slab and $\mathbf{x} = (x, y, z)$ corresponds to the same material point in the deformed configuration. We postulate the following finite deformation $\mathbf{x} = \mathbf{x}(\mathbf{X})$ and temperature field θ:

$$x = X + f(Y), \quad y = Y, \quad z = Z, \tag{1a}$$

$$\theta = \theta(y) = \theta(Y), \tag{1b}$$

where $f(Y)$ is the displacement function to be determined by using a semi-inverse method.

Using Eq. (1a), the deformation gradient $\mathbf{F} = \partial \mathbf{x}/\partial \mathbf{X}$, the left Cauchy–Green strain tensor $\mathbf{B} = \mathbf{F}\mathbf{F}^{\mathrm{T}}$ and its inverse $\mathbf{B}^{-1}$ can be given in the following matrix form:

$$\mathbf{F} = \begin{bmatrix} 1 & f' & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} 1 + (f')^2 & f' & 0 \\ f' & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

$$\mathbf{B}^{-1} = \begin{bmatrix} 1 & -f' & 0 \\ -f' & 1 + (f')^2 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \tag{2}$$

where the prime denotes differentiation with respect to Y. It follows from Eq. (2) that the strain invariants of $\mathbf{B}$, i.e., I_1, I_2, and I_3 can be determined as

$$I_1 = \operatorname{tr} \mathbf{B} = 3 + (f')^2, \quad I_2 = \operatorname{tr} \mathbf{B}^{-1} = 3 + (f')^2,$$

$$I_3 = \det \mathbf{B} = 1. \tag{3}$$

Given $\det \mathbf{F} = 1$, the deformation described by Eq. (1a) is volume preserving and the incompressibility constraint is automatically satisfied.

Because we deal with an inhomogeneous deformation, it is appropriate to define the degree of strain inhomogeneity in the slab. As a measure of the inhomogeneity, we use the following norm that was used by Rajagopal [8]:

$$|||\nabla \mathbf{F}||| := \sqrt{\frac{\partial F_{ij}}{\partial x_k} \frac{\partial F_{ij}}{\partial x_k}} = |f''|. \tag{4}$$

The value of this norm is zero for the isothermal shearing of a slab since the shear strain f' is a constant and does not vary across the thickness of the slab. On the other hand, the value of the norm can be quite high in certain regions of the slab if the strain inhomogeneity is pronounced due to the coupling between stress–strain and temperature fields. In some cases, the strain inhomogeneity can be markedly higher in a layer adjacent to a boundary of the slab and be comparably smaller elsewhere in the slab. This layer is referred to as a boundary layer-like structure (BLS) in this paper.

3. A finite thermoelastic model with entropic origin for the stress

In order to describe the thermomechanics of an isotropic incompressible elastic material, we propose the specific Helmholtz free energy ψ (per unit mass) in the following form:

$$\psi = \frac{\theta}{\rho_0 \theta_0} W(I_1, I_2) + c\left(\theta - \theta_0 - \theta \ln \frac{\theta}{\theta_0}\right), \tag{5}$$

where the parameters c, ρ_0, and θ_0 denote the specific heat capacity, the mass density, and a reference temperature. $W(I_1, I_2)$ can be considered as the value of Helmholtz free energy per unit volume defined at θ_0. It is commonly called the isothermal strain energy function in literature. The specific entropy η and the specific internal energy E can be obtained using the standard thermodynamic relations

$$\eta = -\frac{\partial \psi}{\partial \theta} = -\frac{1}{\rho_0 \theta_0} W(I_1, I_2) + c \ln \frac{\theta}{\theta_0},$$

$$E = \psi + \theta \eta = c(\theta - \theta_0). \tag{6}$$

In Eqs. (5) and (6), c depends on neither the deformation tensor nor the temperature; it is assumed to be a positive-valued constant.

The Cauchy stress $\mathbf{T}$ can be determined from the following equation:

$$\mathbf{T} = -p\mathbf{I} + 2\rho_0 \left(\frac{\partial \psi}{\partial I_1} \mathbf{B} - \frac{\partial \psi}{\partial I_2} \mathbf{B}^{-1} \right), \tag{7}$$

where p is the indeterminate part of the stress due to the incompressibility constraint and called the pressure. The insertion of the Helmholtz free energy given in Eq. (5) into Eq. (7) leads to the following form of stress–strain relation:

$$\mathbf{T} = -p\mathbf{I} + 2\frac{\theta}{\theta_0} \left(\frac{\partial W}{\partial I_1} \mathbf{B} - \frac{\partial W}{\partial I_2} \mathbf{B}^{-1} \right). \tag{8}$$

The relations given in Eqs. (5)–(8) constitute the finite thermoelastic model with entropic origin for the stress, which is a subset of the thermoelastic model developed by Chadwick [2]. As can be seen from Eq. (8), the stress is a linear function of absolute temperature according to our thermoelastic model. The relations $\partial\psi/\partial I_1 = -\theta\partial\eta/\partial I_1$ and $\partial\psi/\partial I_2 = -\theta\partial\eta/\partial I_2$ follow from Eqs. (5) and (6), and imply that at constant temperature the internal energy has no contribution to the stress; thus, the stress is entirely entropic in origin. Uniaxial extension tests performed by Anthony et al. [16] have shown that more than 90% of the tensile stress in a vulcanized rubber can be attributed to the entropic contribution and the rest to the internal energy. Chadwick et al. [2,17] and Allen et al. [18] have considered the dependence of internal energy on the deformation tensor to explain the contribution of the internal energy to the stress. An obvious dependence of stress on the internal energy stems from the thermal expansion of the rubber-like material. However, in this paper, we restrict our treatment to the stress–strain relation given in Eq. (8) that assumes the stress be entropic in origin.

The generalized neo-Hookean model, which has been recently used by several authors in the analysis of inhomogeneous deformations [8,12,13], entails the assumption that the specific heat capacity c of the material is zero. In order to ensure c be a positive constant, Rajagopal [8] points out the need to include a function of the form $C\theta(1 - \ln\theta) + D\theta + E$, where C, D, and E are constants, in the specific Helmholtz free energy ψ. In our thermoelastic model given by Eqs. (5)–(8), we have ensured c be a positive constant by adopting that suggestion. If the interest lies in the determination of spacewise and timewise variation of temperature and motion, c should at least be a positive constant for the slab to store internal energy during the transient period. In fact, c can depend on both the deformation tensor and temperature [2].

3.1. Isothermal strain energy functions used in the thermoelastic model

In this section, we consider the specific forms of the isothermal strain energy function $W(I_1, I_2)$ that can be used in Eq. (8). The previously proposed forms of $W(I_1, I_2)$ contain several material parameters to be determined from curve fitting to the experimental stress–strain data according to Eq. (8). The experiments generally involve some simple deformation modes such as uniaxial extension, biaxial extension, or simple shear at a reference temperature θ_0.

One of the most widely used $W(I_1, I_2)$ has been given by Mooney [14], where

$$W(I_1, I_2) = A_{10}(I_1 - 3) + A_{01}(I_2 - 3), \tag{9}$$

and both A_{10} and A_{01} are the material parameters. The Mooney model represents the uniaxial extension response of rubber-like materials well in the low-to-medium strain domain. It also predicts a constant shear modulus, which equals $2(A_{10} + A_{01})$, and constant normal stress differences in simple shear experiments. The Mooney model generally fails in the large strain domain in simple deformation modes. The prediction of large strain behavior can be accomplished by considering higher-order terms of strain invariants for $W(I_1, I_2)$. Yeoh [15] proposes one such function as

$$W(I_1) = A_{10}(I_1 - 3) + A_{20}(I_1 - 3)^2 + A_{30}(I_1 - 3)^3, \tag{10}$$

where A_{10}, A_{20}, and A_{30} are the material parameters. Experimental investigations [19,20] have revealed that at higher strains, the tangent modulus of rubber-like materials increases with increasing strain. The experiments by Seki et al. [21] and Yeoh [22] have shown that the shear modulus of a rubber is nearly constant in the low-to-medium shear strain domain, but increases with strain in the large strain domain. Although Eq. (10) contains terms only in I_1, it represents the response of rubber-like materials in the large strain domain fairly well because it can adapt the experimentally observed stiffening of the rubber-like materials.

To estimate the material parameters in Eqs. (9) and (10), we use the isothermal simple shear data of Seki et al. [21] for a vulcanized rubber at room temperature ($\theta_0 = 296\,\mathrm{K}$). The shear modulus $2(A_{10} + A_{01})$ is estimated to be 500 kPa by the Mooney model. Since we are not interested in the complete rheological characterization of the rubber using the Mooney model, the relative values of A_{10} and A_{01} are set based on the experimentally reported relation $A_{10} > A_{01}$ [20]. The Mooney model with the set parameters $A_{10} = 200\,\text{kPa}$ and $A_{01} = 50\,\text{kPa}$ predicts the linear portion of the shear stress–strain curve, whereas the Yeoh model with the parameters $A_{10} = 248\,\text{kPa}$, $A_{20} = -1.484\,\text{kPa}$, and $A_{30} = 0.302\,\text{kPa}$ predicts both the linear and non-linear portions. At higher shear strains (above 1.28), the Yeoh model predicts the increase in shear modulus with increasing strain. In this paper, the stiffening of the rubber-like materials with increasing shear strain is referred to as shear stiffening. It should be noted that the thermoelastic model in Eq. (8) suggests a linear increase in modulus with increasing temperature, which is independent of the chosen form of $W(I_1, I_2)$. In this paper, this is referred to as temperature stiffening.

When the isothermal strain energy functions given in Eqs. (9) and (10) are inserted into Eq. (8), we obtain the stress–strain relations for the generalized Mooney and the Yeoh models. For the sake of brevity, the slabs whose stress–strain behavior are given by Eqs. (8) and (9) and by Eqs. (8) and (10) are called the Mooney slab and the Yeoh slab, respectively.

4. Field equations

The Mooney and the Yeoh slabs under a temperature gradient undergo the inhomogeneous shear deformation described in Section 2. The explicit form of the stress–strain and temperature fields can be obtained by using a semi-inverse method and satisfying the linear momentum balance equations and the energy balance equation.

4.1. Determination of the temperature field

Let us consider the energy balance equation for an incompressible slab

$$\rho_0 \dot{\mathbf{E}} - \mathbf{T} : \mathbf{d} + \operatorname{div} \mathbf{q} - \rho_0 r = 0. \tag{11}$$

In the above equation, $\mathbf{d}$, $\mathbf{q}$, and r denote the symmetric part of the velocity gradient, the heat flux vector, and the radiant heating, respectively. The first term represents the material (substantial) derivative of E. The tensorial product $\mathbf{T} : \mathbf{d}$ is known as stress power. The third term represents heat conduction. In this paper, we are concerned with steady-state heat conduction in a stationary slab with a constant thermal conductivity k when no radiant heating is present. Hence, using the postulated temperature field in Eq. (1b) and Fourier's law of heat conduction to describe $\mathbf{q}$, Eq. (11) reduces to the following boundary value problem with the prescribed temperatures

$$\operatorname{div} \mathbf{q} = \frac{\mathrm{d}}{\mathrm{d}y}\left(-k \frac{\mathrm{d}\theta}{\mathrm{d}y} \right) = -k\theta'' = 0,$$

$$\theta(0) = \theta_{\mathrm{C}} \quad \text{and} \quad \theta(H) = \theta_{\mathrm{H}}. \tag{12}$$

The slab always has a colder boundary ($Y = 0$) whose temperature is $\theta_{\mathrm{C}} = 298$ K. The boundary at $Y = H$ is kept at various higher temperatures θ_{H} to determine the effects of the temperature difference or gradient on the inhomogeneous shear deformation. The energy balance described in Eq. (12) is decoupled from the strain field. Thus, the temperature field for both the Mooney and the Yeoh slabs is the same and can be obtained analytically from Eq. (12)

$$\theta = \theta(Y) = \theta_{\mathrm{C}} + \frac{\theta_{\mathrm{H}} - \theta_{\mathrm{C}}}{H} Y. \tag{13}$$

Eq. (13) shows that the temperature increases linearly from the colder boundary ($Y = 0$) to the hotter boundary ($Y = H$). In this paper, the dimensional form of the temperature and the stress field is used because we want to gain a physical intuition by obtaining explicit temperature dependence of the shear strain and the stress field. Additionally, making these variables dimensionless does not significantly simplify the problem formulation and solution.

4.2. Determination of the stress–strain fields

The balance of linear momentum for a stationary slab with no body force can be recorded as

$$\operatorname{div} \mathbf{T} = \mathbf{0}. \tag{14}$$

Without loss of generality, in order to simplify the algebraic manipulations we define a modified pressure Π

$$\Pi = p - 2\frac{\theta}{\theta_0}\left\{\frac{\partial W}{\partial I_1} - [1 + (f')^2]\frac{\partial W}{\partial I_2}\right\}. \tag{15}$$

After substituting Eq. (2) into Eq. (8) and making use of Eq. (15), $\mathbf{T}$ is plugged into Eq. (14) and using the postulated deformation field in Eq. (1a), we obtain

$$-\frac{\partial \Pi}{\partial X} + \frac{2}{\theta_0}\frac{\mathrm{d}}{\mathrm{d}Y}\left[\theta\left(\frac{\partial W}{\partial I_1} + \frac{\partial W}{\partial I_2}\right)f'\right] = 0, \tag{16a}$$

$$f'\frac{\partial \Pi}{\partial X} - \frac{\partial \Pi}{\partial Y} = 0, \tag{16b}$$

$$-\frac{\partial \Pi}{\partial Z} = 0. \tag{16c}$$

It follows from Eq. (16c) that $\Pi = \Pi(X, Y)$, hence the integration of Π over X in Eq. (16a) yields the following form of Π:

$$\Pi = \Pi(X, Y) = \alpha X + \beta, \tag{17}$$

where α and β are functions of Y. Inserting Eq. (17) into Eq. (16b), we determine α, β, Π, and p as

$$\alpha = \alpha(Y) = A, \quad \beta = \beta(Y) = Af + B,$$

$$\Pi = \Pi(X, Y) = A(X + f) + B, \tag{18a}$$

$$p = p(X, Y) = A(X + f) + B + 2\frac{\theta}{\theta_0}\left\{\frac{\partial W}{\partial I_1} - [1 + (f')^2]\frac{\partial W}{\partial I_2}\right\}, \tag{18b}$$

where A denotes the pressure gradient along the x-direction, i.e., $\partial p/\partial x = \partial p/\partial X$, and B denotes an arbitrary pressure constant.

From this point forward, we assume $A = 0$, which enforces p to have finite values at $\pm\infty$; the deformation becomes akin to Couette flow of fluids. Hence, Eq. (16a) turns into the following ordinary differential equation with the prescribed displacement boundary conditions:

$$\left[\theta\left(\frac{\partial W}{\partial I_1} + \frac{\partial W}{\partial I_2}\right)f'\right]' = 0,$$

$$f(0) = 0 \quad \text{and} \quad f(H) = U, \tag{19}$$

in which U is the prescribed displacement of the hotter boundary ($Y = H$), while the colder boundary ($Y = 0$) is fixed to a rigid surface. Eq. (19) can be solved after the specification of the slab (Mooney or Yeoh slab) by using Eq. (9) or (10) and the invariants I_1, I_2 given in Eq. (3). We solve Eq. (19) numerically for the displacement function f and the shear strain f' using a finite difference method as will be explained in Section 5. Once f' is determined, we use Eq. (8) to obtain the stress field

$$p = p(Y) = B + 2\frac{\theta}{\theta_0}\left\{\frac{\partial W}{\partial I_1} - [1 + (f')^2]\frac{\partial W}{\partial I_2}\right\}, \tag{20a}$$

$$T_{xx} = -B + 2\frac{\theta}{\theta_0}\left(\frac{\partial W}{\partial I_1} + \frac{\partial W}{\partial I_2}\right)(f')^2, \tag{20b}$$

$$T_{xy} = 2\frac{\theta}{\theta_0}\left(\frac{\partial W}{\partial I_1} + \frac{\partial W}{\partial I_2}\right)f', \tag{20c}$$

$$T_{yy} = -B, \tag{20d}$$

$$T_{zz} = -B + 2\frac{\theta}{\theta_0}\frac{\partial W}{\partial I_2}(f')^2, \tag{20e}$$

$$T_{xz} = T_{yz} = 0. \tag{20f}$$

In order to eliminate the constant B, the first normal stress difference N_1 and the second normal stress difference N_2 are defined as

$$N_1 = T_{xx} - T_{yy} = 2\frac{\theta}{\theta_0}\left(\frac{\partial W}{\partial I_1} + \frac{\partial W}{\partial I_2}\right)(f')^2, \tag{21a}$$

$$N_2 = T_{zz} - T_{yy} = 2\frac{\theta}{\theta_0}\frac{\partial W}{\partial I_2}(f')^2. \tag{21b}$$

4.3. Determination of principal stresses and principal directions

The principal stresses $T_{\mathrm{p}}^{(i)}$ and the corresponding principal directions $\mathbf{n}^{(i)}$ (the unit normal of the principal plane for $i = 1, 2, 3$) are obtained by solving the following eigenvalue problem:

$$(\mathbf{T} - T_{\mathrm{p}}^{(i)}\mathbf{I})\cdot\mathbf{n}^{(i)} = \mathbf{0}. \tag{22}$$

Here, we adopt the convention $T_{\mathrm{p}}^{(1)} > T_{\mathrm{p}}^{(2)} > T_{\mathrm{p}}^{(3)}$. It follows from Eqs. (20) and (22) that the principal stresses are

$$T_{\mathrm{p}}^{(1)} = -B + \frac{\theta}{\theta_0}\left(\frac{\partial W}{\partial I_1} + \frac{\partial W}{\partial I_2}\right) \times [f' + \sqrt{4 + (f')^2}]f', \tag{23a}$$

$$T_{\mathrm{p}}^{(2)} = -B + 2\frac{\theta}{\theta_0}\frac{\partial W}{\partial I_2}(f')^2, \tag{23b}$$

$$T_{\mathrm{p}}^{(3)} = -B + \frac{\theta}{\theta_0}\left(\frac{\partial W}{\partial I_1} + \frac{\partial W}{\partial I_2}\right) \times [f' - \sqrt{4 + (f')^2}]f', \tag{23c}$$

and the corresponding principal directions are

$$\mathbf{n}^{(1)} = \frac{1}{\sqrt{8 + 2(f')^2 - 2f'\sqrt{4 + (f')^2}}} \times \{2, -(f' - \sqrt{4 + (f')^2}), 0\}, \tag{24a}$$

$$\mathbf{n}^{(2)} = \{0, 0, 1\}, \tag{24b}$$

$$\mathbf{n}^{(3)} = \frac{1}{\sqrt{8 + 2(f')^2 + 2f'\sqrt{4 + (f')^2}}} \times \{2, -(f' + \sqrt{4 + (f')^2}), 0\}. \tag{24c}$$

5. Numerical solutions for Mooney and Yeoh slabs

Any inhomogeneous shear deformation should, in principle, be studied within the context of a subclass of isotropic elastic materials. In Section 4, we have developed the field equations within the context of the finite thermoelastic model with entropic origin for the stress; however, the isothermal strain energy function $W(I_1, I_2)$ has not been explicitly specified. We can incorporate any suitable $W(I_1, I_2)$ obtained from the isothermal response of the rubber-like slab (see Section 3.1). Here, we consider a Mooney slab and a Yeoh slab whose strain energy functions are given in Eqs. (9) and (10), respectively. Then, the linear momentum balance equation (19) is numerically solved for f.

5.1. Numerical solution of the linear momentum balance equation

After inserting Eqs. (9) and (13) into Eq. (19), the linear momentum balance equation for the Mooney slab becomes

$$f'' + \frac{\theta_{\mathrm{H}} - \theta_{\mathrm{C}}}{\theta_{\mathrm{C}}H + (\theta_{\mathrm{H}} - \theta_{\mathrm{C}})Y}f' = 0,$$

$$f(0) = 0 \quad \text{and} \quad f(H) = U. \tag{25}$$

We deduce from Eq. (25) that the material parameters A_{10} and A_{01} of the Mooney slab do not affect the strain field. However, A_{10} and A_{01} do influence the stress field and the principal stresses, as can be seen in Section 5.2. Furthermore, Eq. (25) can be obtained for a neo-Hookean slab with $A_{10} = \text{constant}$ and $A_{01} = 0$. Thus, the strain field for a Mooney slab and a neo-Hookean slab is identical although the stress field can differ.

The incorporation of Eqs. (3), (10), and (13) into Eq. (19) yields the following linear momentum balance equation for the Yeoh slab:

$$[A_{10} + 6A_{20}(f')^2 + 15A_{30}(f')^4]f'' + \frac{\theta_{\mathrm{H}} - \theta_{\mathrm{C}}}{\theta_{\mathrm{C}}H + (\theta_{\mathrm{H}} - \theta_{\mathrm{C}})Y}[A_{10} + 2A_{20}(f')^2 + 3A_{30}(f')^4]f' = 0,$$

$$f(0) = 0 \quad \text{and} \quad f(H) = U. \tag{26}$$

We consider a slab with a thickness of $H = 0.01\,\mathrm{m}$. The slab is sheared by an amount $U = 0.04\,\mathrm{m}$. Eq. (25) is a linear ordinary differential equation and can be solved analytically. However, Eq. (26) is non-linear and, in general, solving it requires a numerical method. We choose to use a finite difference method to solve Eqs. (25) and (26) for f. We divide the physical domain along the Y direction $[0, H]$ into $N - 1$ intervals with sizes

$\Delta Y = 10^{-4}$ m (or $N = 101$). The derivatives in Eqs. (25) and (26) are approximated using the discrete values of f, i.e., f_i in the following second-order accurate finite difference equations:

$$f_1' = (-3f_1 + 4f_2 - f_3)/2\Delta Y \text{ and}$$

$$f_1'' = (2f_1 - 5f_2 + 4f_3 - f_4)/(\Delta Y)^2 \quad \text{at } i = 1,$$

$$f_N' = (3f_N - 4f_{N-1} + f_{N-2})/2\Delta Y \text{ and}$$

$$f_N'' = (2f_N - 5f_{N-1} + 4f_{N-2} - f_{N-3})/(\Delta Y)^2 \quad \text{at } i = N,$$

$$f_i' = (f_{i+1} - f_{i-1})/2\Delta Y \text{ and}$$

$$f_i'' = (f_{i+1} - 2f_i + f_{i-1})/(\Delta Y)^2 \quad \text{at } i = 2, N-1. \tag{27}$$

The substitution of Eq. (27) into Eqs. (25) and (26) for the inner-domain points ($i = 2, N-1$) yields a set of $N-2$ linear algebraic equations for the Mooney slab and a set of $N-2$ non-linear algebraic equations for the Yeoh slab, respectively. We recognize that $f_1 = 0$ and $f_N = U$ from the boundary conditions. The linear equations for the Mooney slab are solved directly using the LU-factorization technique. The non-linear equations for the Yeoh slab are iteratively solved using Newton's method. In each iteration, the non-linear equations are first linearized locally around an initial guess vector for f_i. The Jacobian matrix, whose components contain derivatives with respect to the unknowns f_i, is evaluated using a forward difference method with $\Delta f_i = 10^{-8}$ m. Then, the resulting linear equations are solved using the LU-factorization technique. The initial guess vector for f_i is updated using the newly computed values of f_i; and the iterations continue with the new guess vector. The convergence criterion for the Newton's method is based on the absolute value of the difference of the solution vector over two successive iterations with respect to the previously computed solution. The iterations are stopped when this criterion falls below 10^{-10}. No significant change in the solution is observed when the number of grid points N is increased from 101 to 151. Therefore, we assume that a grid-independent solution is attained for $N = 101$ or $\Delta Y = 10^{-4}$ m.

5.2. *Solutions to the stress field, principal stresses, and principal directions*

Having determined f numerically, we obtain f' and f'' from Eq. (27) and the degree of strain inhomogeneity from Eq. (4) for each slab. Inserting Eqs. (9) and (13) into Eq. (20c), we determine the shear stress T_{xy} for the Mooney slab

$$T_{xy} = 2\left(\frac{\theta_C}{\theta_0} + \frac{\theta_H - \theta_C}{\theta_0 H} Y\right)(A_{10} + A_{01}) f'. \tag{28}$$

Similarly, the incorporation of Eqs. (3), (10), and (13) into Eq. (20c) yields T_{xy} for the Yeoh slab

$$T_{xy} = 2\left(\frac{\theta_C}{\theta_0} + \frac{\theta_H - \theta_C}{\theta_0 H} Y\right) \times [A_{10} + 2A_{20}(f')^2 + 3A_{30}(f')^4] f'. \tag{29}$$

It follows from Eqs. (9), (13), and (21) that the first normal stress difference N_1 and the second normal stress difference N_2 for the Mooney slab are

$$N_1 = 2\left(\frac{\theta_C}{\theta_0} + \frac{\theta_H - \theta_C}{\theta_0 H} Y\right)(A_{10} + A_{01})(f')^2, \tag{30a}$$

$$N_2 = 2\left(\frac{\theta_C}{\theta_0} + \frac{\theta_H - \theta_C}{\theta_0 H} Y\right) A_{01} (f')^2. \tag{30b}$$

Eq. (30) shows that N_2 is smaller than N_1 in the Mooney slab. N_1 and N_2 for the Yeoh slab are obtained by making use of Eqs. (3), (10), (13), and (21)

$$N_1 = 2\left(\frac{\theta_C}{\theta_0} + \frac{\theta_H - \theta_C}{\theta_0 H} Y\right)[A_{10} + 2A_{20}(f')^2 + 3A_{30}(f')^4](f')^2, \tag{31a}$$

$$N_2 = 0. \tag{31b}$$

Using Eqs. (9), (13), and (23a), the major principal stress $T_p^{(1)}$ for the Mooney slab can be given within the pressure constant B as

$$T_p^{(1)} + B = \left(\frac{\theta_C}{\theta_0} + \frac{\theta_H - \theta_C}{\theta_0 H} Y\right)(A_{10} + A_{01}) \times [f' + \sqrt{4 + (f')^2}] f'. \tag{32}$$

Similarly, using Eqs. (3), (10), (13), and (23a), for the Yeoh slab we obtain

$$T_{\mathrm{p}}^{(1)} + B = \left(\frac{\theta_{\mathrm{C}}}{\theta_0} + \frac{\theta_{\mathrm{H}} - \theta_{\mathrm{C}}}{\theta_0 H} Y\right)[A_{10} + 2A_{20}(f')^2 + 3A_{30}(f')^4][f' + \sqrt{4 + (f')^2}]f'. \tag{33}$$

The angle that the normal vector of the major principal plane makes with the x-direction, denoted by ϕ, for both the Mooney and the Yeoh slabs is determined from Eq. (24a) as

$$\phi = \arccos(\mathbf{n}^{(1)} \cdot \mathbf{e}_x) = \arccos\left(\frac{2}{\sqrt{8 + 2(f')^2 - 2f'\sqrt{4 + (f')^2}}}\right). \tag{34}$$

It should be noted from Eq. (34) that ϕ depends only on f' explicitly for both the Mooney and the Yeoh slabs. The material parameters of the Mooney slab do not affect ϕ, whereas those of the Yeoh slab affect ϕ implicitly through their influence on f'.

6. Results and discussion

There exists a markedly high f' near the colder boundary ($Y = 0$) in the Mooney slab, as seen in Fig. 1. The shear strain under isothermal conditions is well-known to be $f' = U/H$ that is equal to 4 for both the Mooney and the Yeoh slabs. The deviation of the shear strain from the isothermal shear solution is 65% at the colder boundary for $\theta_{\mathrm{H}} = 773$ K. Fig. 1 also shows that f' gets smaller from the colder to the hotter boundary. This can be explained by the temperature stiffening of the Mooney slab. When the slab is hotter at a point along its thickness, its modulus is higher at that point and consequently f' is smaller. As θ_{H} and consequently the temperature difference across the thickness increase, the variation of f' and consequently its deviation from the isothermal solution also increase. As seen in Fig. 2, f' is greater at the colder boundary and smaller at the hotter boundary of the Yeoh slab. The variation of f' across the thickness increases with an increasing temperature difference. The effect of temperature difference is obviously similar for both the Mooney and the Yeoh slabs because the stress–strain relation of the finite thermoelastic model governs the temperature stiffening of both slabs, as can be seen from Eq. (8). On the other hand, at the same temperature difference or at the same θ_{H}, the variation of f' is smaller in the Yeoh slab than in the Mooney slab (see Figs. 1 and 2). Hence, the shear deformation is more homogeneous in the Yeoh slab than in the Mooney

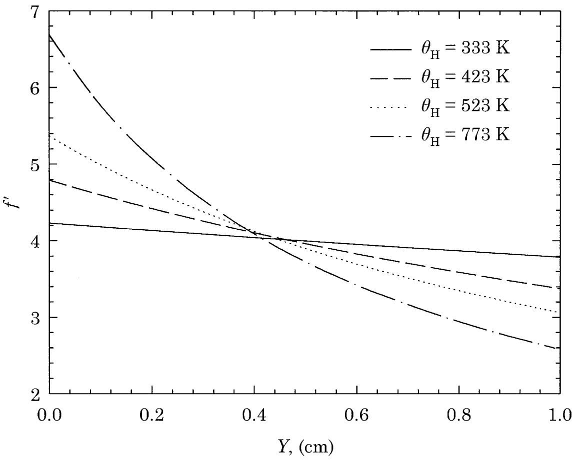

Fig. 1. Variation of the shear strain f' across the thickness of the Mooney slab for various hotter boundary temperatures θ_{H}.

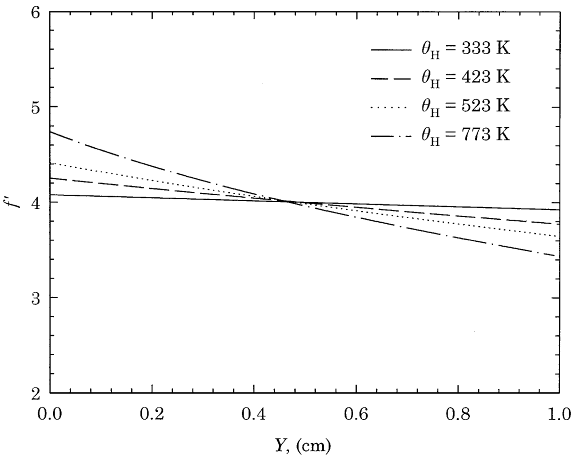

Fig. 2. Variation of the shear strain f' across the thickness of the Yeoh slab for various hotter boundary temperatures θ_{H}.

slab. This can be explained by the difference in the isothermal strain energy functions of the slabs: the Mooney slab has a constant shear modulus, whereas the Yeoh slab has an increasing modulus with increasing shear strain (shear stiffening). Hence, the shear stiffening of the Yeoh slab tends to reduce the variation of f'.

A boundary layer-like structure BLS can be defined as a layer of the slab where f' is much higher than elsewhere in the slab. Although the formation of a BLS could be determined by inspecting the values of f', a more definitive method is to analyze the values of the norm defined in Eq. (4). This norm is equal to the absolute value of the shear strain gradient $|f''|$. Fig. 3 illustrates that a BLS seems to form near the colder boundary of the Mooney slab, which is indicated by much higher values of $|f''|$ near the colder boundary than in the rest of the slab. As the temperature difference across the thickness increases, the formation of a BLS near the colder boundary becomes more apparent. It should be mentioned that although theoretically a BLS seems to appear in the Mooney slab, the temperatures required for that are unrealistically high (e.g. $\theta_H = 773\,K$). It is well known in rubber chemistry literature that even the most temperature-resistant rubbers begin to degrade at temperatures above 673 K [23]. The tendency to form a BLS near the colder boundary is much weaker for the

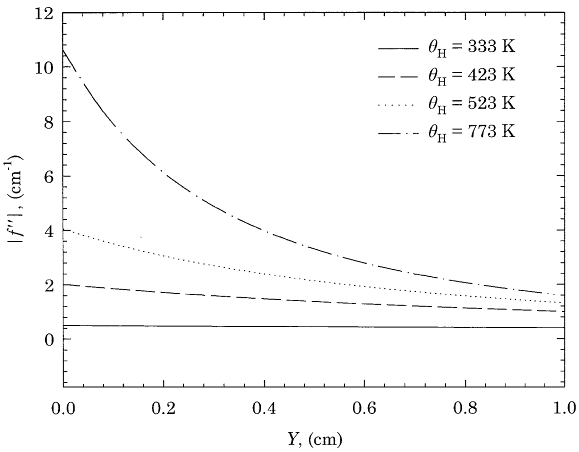

Fig. 3. Variation of the absolute value of the shear strain gradient $|f''|$ across the thickness of the Mooney slab for various hotter boundary temperatures θ_H.

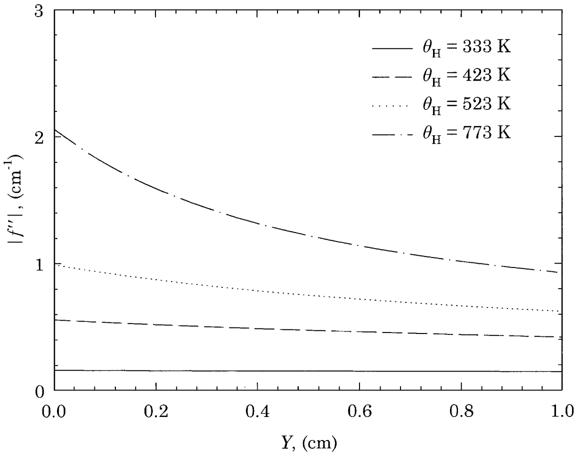

Fig. 4. Variation of the absolute value of the shear strain gradient $|f''|$ across the thickness of the Yeoh slab for various hotter boundary temperatures θ_H.

Yeoh slab, as shown in Fig. 4 by the smaller values of $|f''|$ and smaller variation of $|f''|$ across the thickness compared with those of the Mooney slab. There is no tendency to form a BLS even at the highest temperature difference ($\theta_H = 773\,K$ case). The difference between the two slabs arises from the previously explained shear stiffening of the Yeoh slab. The shear stiffening of the Yeoh slab competes with the temperature stiffening resulting in both a smaller $|f''|$ and its variation across the thickness. The annihilation of BLS is not observed in the Mooney slab, which has constant shear modulus, as there is no competition between the temperature stiffening and shear stiffening.

Fig. 5 shows that T_{xy} is constant across the thickness, and increases as the temperature difference increases for the Mooney slab, even though f' varies across the thickness. Fig. 6 shows a similar pattern of T_{xy} in the Yeoh slab. The satisfaction of the linear momentum balance equation (19) leads to the constancy of T_{xy} at a given temperature difference. T_{xy} is higher in the Yeoh slab than in the Mooney slab at the same temperature difference due to the shear stiffening and, consequently, higher modulus of the Yeoh slab.

Fig. 7 illustrates that N_1 in the Mooney slab increases from the hotter to the colder boundary and becomes very high near the colder boundary. Both N_1 and its variation increase as the

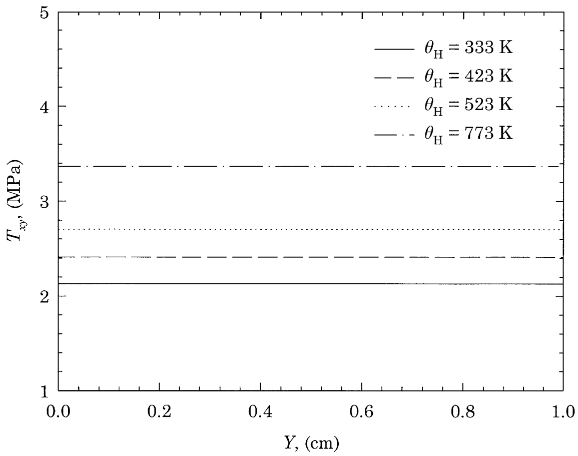

Fig. 5. Variation of the shear stress T_{xy} across the thickness of the Mooney slab for various hotter boundary temperatures θ_H.

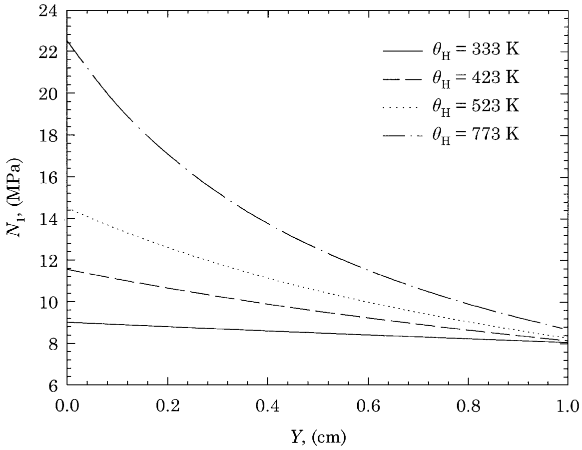

Fig. 7. Variation of the first normal stress difference N_1 across the thickness of the Mooney slab for various hotter boundary temperatures θ_H.

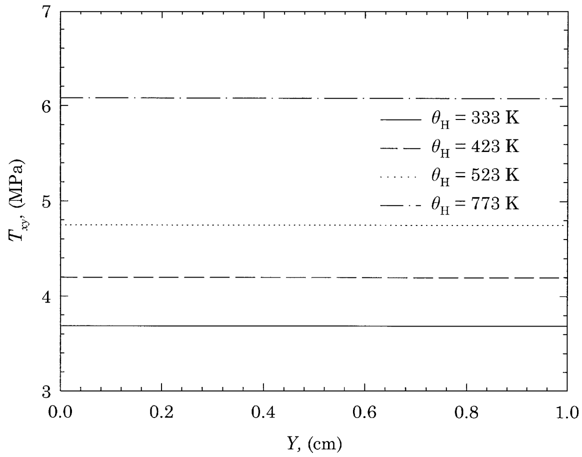

Fig. 6. Variation of the shear stress T_{xy} across the thickness of the Yeoh slab for various hotter boundary temperatures θ_H.

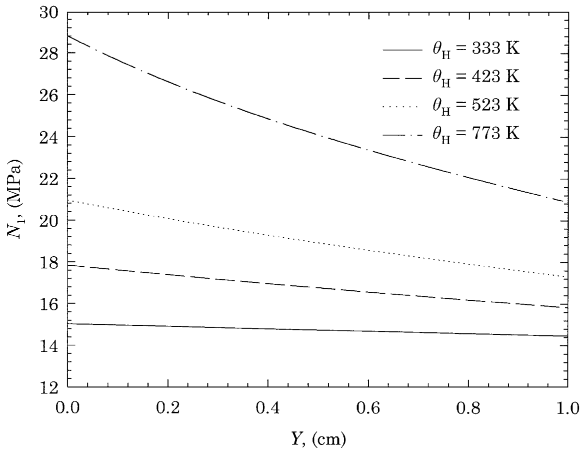

Fig. 8. Variation of the first normal stress difference N_1 across the thickness of the Yeoh slab for various hotter boundary temperatures θ_H.

temperature difference across the thickness increases. This behavior is very similar to that observed for the variation of f'. Near the colder boundary of the Mooney slab at $\theta_H = 773\,K$, not only does a BLS seem to appear, but also N_1 becomes much greater than in the rest of the slab. As illustrated in Fig. 8, N_1 in the Yeoh slab increases from the hotter to the colder boundary. It also increases with increasing temperature difference as in the Mooney slab case. However, the variation of N_1 at the same temperature difference is smaller in the Yeoh slab than in the Mooney slab. This is in accordance with the fact that the variation of f' is smaller and the tendency to form a BLS is much weaker in the Yeoh slab than in the Mooney slab. Hence, the shear stiffening of the Yeoh slab reduces not only the formation of a BLS, but also the variation of N_1. N_1 is greater in the Yeoh slab than in the Mooney slab at the same temperature difference, which can also be attributed to the above-mentioned shear stiffening of the Yeoh slab.

N_1 is generally much greater than T_{xy} for both the Mooney and the Yeoh slabs, as can be seen from Figs. 5 and 7 and from Figs. 6 and 8, respectively. In other words, although the slabs are sheared, the stress field is dominated by the very large N_1. This resulted from the non-linear stress–strain behavior of the slabs. We recognize from Eqs. (28)–(31) that the ratio of N_1 to T_{xy} is equal to f'. T_{xy} would be dominant at infinitesimal f', while N_1 approaches zero. On the other hand, N_1 surpasses T_{xy} at large f'.

The major principal stress is determined within a pressure constant B and is denoted as $T_{\mathrm{p}}^{(1)} + B$. Fig. 9 illustrates that $T_{\mathrm{p}}^{(1)} + B$ varies across the thickness, and is greater at the colder boundary of the Mooney slab. It also increases with increasing temperature difference. The same behavior is observed in the Yeoh slab, as seen in Fig. 10. It is noted that $T_{\mathrm{p}}^{(1)} + B$ is almost equal to N_1 for both slabs. This can be attributed to the fact that $T_{\mathrm{p}}^{(1)} + B$ receives most of its contribution from N_1 that is much greater than T_{xy} at large strains. Although $T_{\mathrm{p}}^{(1)} + B$ at the same temperature difference is greater in the Yeoh slab than in the Mooney slab, its variation is smaller in the Yeoh slab. This behavior is also observed when the values of N_1 and its variation are considered. All of these can be explained by the shear stiffening of the Yeoh slab, which increases the modulus of the slab and

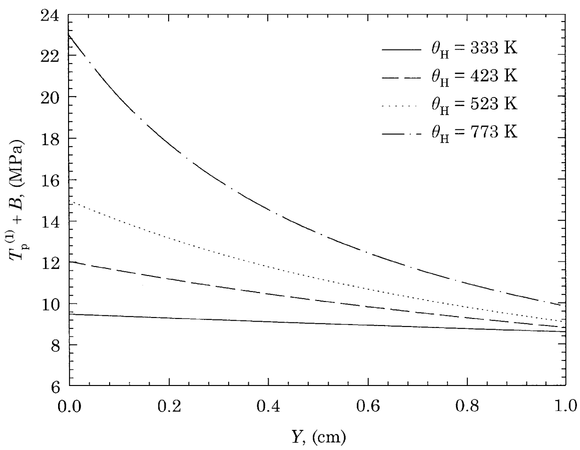

Fig. 9. Variation of the major principal stress within the pressure constant B, i.e., $T_{\mathrm{p}}^{(1)} + B$ across the thickness of the Mooney slab for various hotter boundary temperatures θ_{H}.

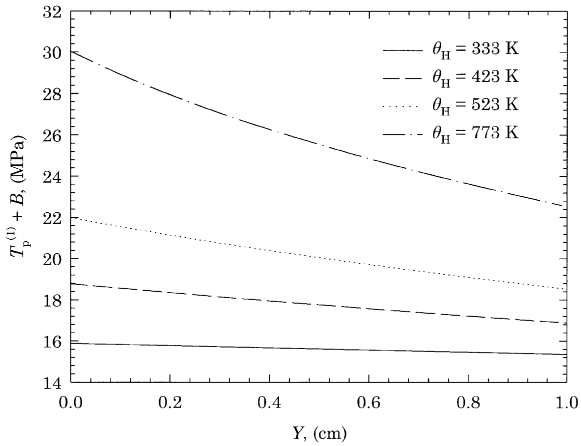

Fig. 10. Variation of the major principal stress within the pressure constant B, i.e., $T_{\mathrm{p}}^{(1)} + B$ across the thickness of the Yeoh slab for various hotter boundary temperatures θ_{H}.

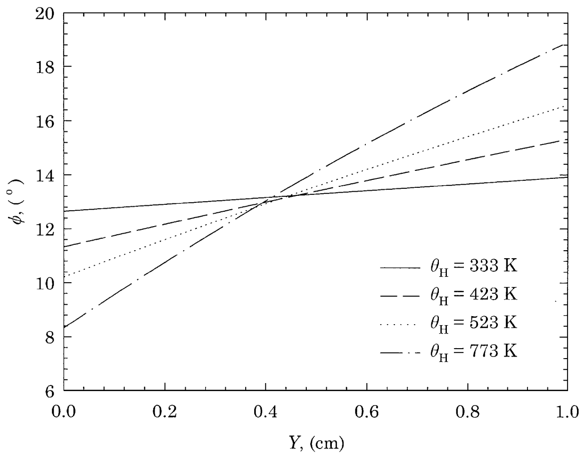

Fig. 11. Variation of the angle ϕ across the thickness of the Mooney slab for various hotter boundary temperatures θ_{H}.

results in greater T_{xy}, N_1, and $T_{\mathrm{p}}^{(1)} + B$ compared with those in the Mooney slab. However, the shear stiffening of the Yeoh slab is counteracted by the temperature stiffening, which causes a reduction in the variation of f', an annihilation of the BLS formation, and consequently smaller variations of N_1 and $T_{\mathrm{p}}^{(1)} + B$.

The variation of ϕ — the angle that the normal vector of the major principal plane makes with the x-axis — for the Mooney slab is shown in Fig. 11.

At a given temperature difference, ϕ increases from the colder boundary to the hotter boundary. The variation of ϕ increases with increasing temperature difference. Fig. 12 illustrates a similar behavior of ϕ in the Yeoh slab. However, the variation of ϕ is smaller in the Yeoh slab than in the Mooney slab, which is in line with the smaller variation of f' due to the shear stiffening of the Yeoh slab.

Let us make some further comments on the calculated stress field, the principal stresses, and principal directions. In linearized elasticity, if a material is sheared, there exist only a shear stress T_{xy} and zero normal stress differences; the major principal stress equals T_{xy}. The normal vector of the major principal plane forms an angle of 45° with the x-axis. In the non-linear thermoelastic materials under consideration, we obtain very large first normal stress difference N_1 and major principal stress $T_{\mathrm{p}}^{(1)} + B$ compared with the shear stress T_{xy}. The determination of principal stresses and principal directions is important to the understanding of possible crack propagation at large strains. Let us assume that a pre-existing crack has the critical size to propagate. What will be the direction of propagation? Usually, one considers that the orientation presenting the highest risk of fracture is perpendicular to the largest in-plane principal stress (or major principal stress) [24,25], and we assume that

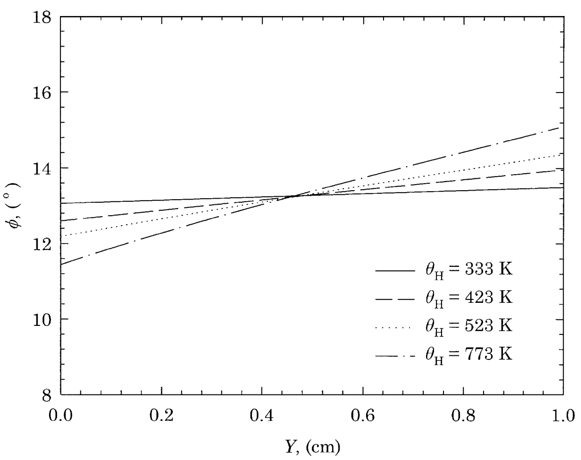

Fig. 12. Variation of the angle ϕ across the thickness of the Yeoh slab for various hotter boundary temperatures θ_{H}.

this can be extended to the rubber-like materials. From this perspective, we can propose that the propagation might occur in the most unfavorable mode, Mode I, i.e., the crack opening under the effect of $T_{\mathrm{p}}^{(1)} + B$ on a plane whose normal makes an angle ϕ with the x-axis. Very small values of ϕ (much less than 45°) and high values of $T_{\mathrm{p}}^{(1)} + B$ at large f' imply that the direction of possible crack propagation in the slab could be almost parallel to the thickness direction. This implication is quite interesting since the direction of crack propagation would be parallel to the thickness direction if the rubber-like slab experiences a large tensile stress in a uniaxial extension experiment. Considering that N_1 contributes to $T_{\mathrm{p}}^{(1)} + B$ much more than T_{xy} at large f', we can suggest that the normal stresses might play significant roles in the crack propagation.

7. Conclusions

The temperature and the stress–strain fields are determined in a generalized Mooney slab and a generalized Yeoh slab that are subjected to simultaneous shearing and a temperature difference across their thickness. The Mooney slab with a high temperature difference across its thickness seems to exhibit a boundary layer-like structure near its colder boundary where the shear strain is much greater than in the rest of the slab. On the other hand, the shear stiffening of the Yeoh slab is counteracted by the temperature stiffening, thus reducing the variation of the shear strain and the formation of a boundary layer-like structure. The variation of the first normal stress difference and the major principal stress is found to be very similar to the variation of the shear strain in both slabs. Hence, the existence of a boundary layer-like structure for the first normal stress difference and the major principal stress can be associated with the existence of a boundary layer-like structure for shear strain.

Both the Mooney and the Yeoh slabs experience much greater first normal stress difference than the shear stress. The major principal stress is dominated by the first normal stress difference rather than the shear stress, although the slabs are sheared.

Consequently, the normal vector of the major principal plane forms a much smaller angle than 45° with the x-axis. This non-linear phenomenon can be instrumental in possible Mode I type crack propagation in the slabs.

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